MIRB Mathematical Information Retrieval Benchmark (2505.15585v1)

Abstract

MIRB Mathematical Information Retrieval Benchmark (2505.15585v1), p.1

> Mathematical Information Retrieval (MIR) is the task of retrieving information from mathematical documents and plays a key role in various applications, including theorem search in mathematical libraries, answer retrieval on math forums, and premise selection in automated theorem proving

The benchmark focuses on mathematical information retrieval.

MIRB Mathematical Information Retrieval Benchmark (2505.15585v1), p.1

> MIRB includes four tasks—semantic statement retrieval, question-answer retrieval, premise retrieval, and formula retrieval—spanning a total of 12 datasets

4 tasks of the benchmark

Conclusion and Limitation

MIRB Mathematical Information Retrieval Benchmark (2505.15585v1), p.9

> hese tasks span both semantic-based and reasoning-based retrieval settings. We evaluate 13 retrieval models and observe that while their performance on semantic-based retrieval is moderate, they perform poorly on reasoning-based tasks

Main finding

Tables and figures

Not very interesting.

Introduction

MIRB Mathematical Information Retrieval Benchmark (2505.15585v1), p.1

> n this case, the MIR query can be either a natural language or formal statement, and the corpus consists of declarations in mathlib4. Another example is students searching for similar questions or answers on Mathematics Stack Exchange to help them solve problems. Here, the user's question serves as the query, and the corpus includes all question and answer posts on the forum

Queries for mathematics is natural language.

Method

Semantic Retrieval

- Used in the semantic retrieval of theorems and duplicate questions.

QA Retrieval

- Retrieve relevant answers or proofs.
- Challenge is understanding the mathematical intent of the question.

Premise Retrieval

- Retrieve definitions, theorems and lemmas that are useful for proving target theorem or advancing the current proof state.

Formula Retrieval

- Retrieve mathematical expressions that are relevant to a given query formula, optionally including the models context.

Results

MIRB Mathematical Information Retrieval Benchmark (2505.15585v1), p.8

> mong the evaluated tasks, models generally perform better on semantic retrieval tasks such as Semantic Statement Retrieval and Formula Retrieval, while their performance degrades on reasoning-oriented tasks, especially Premise Retrieval

Performance degrades on reasoning tasks. Models are not able to link the queried mathematical expression and surrounding context.

MIRB Mathematical Information Retrieval Benchmark (2505.15585v1), p.8

> For formal premise retrieval datasets like LeanDojo, MAPL, and HolStep, embedding models often struggle because they are not extensively pre-trained on large corpora of formal language data

Another failure mode - lack of relevant training data.